

CONTEXT-ADAPTIVE AI ASSERTIVENESS DOESN'T BEAT FIXED PERSONAS: A NEGATIVE RESULT IN HUMAN-AI COLLABORATION

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ABSTRACT

Effective human-AI collaboration requires balancing AI assistance with human agency. Recent work shows contrarian AI improves critical thinking but reduces satisfaction, while supportive AI maintains comfort but risks overreliance (??). We hypothesized that dynamically adapting AI assertiveness based on real-time cognitive load would outperform fixed interaction styles. Through simulation experiments modeling human decision-making with neural networks, we tested three conditions: always-supportive AI, always-challenging AI, and adaptive AI that switches modes based on cognitive load proxies. Contrary to our hypothesis, adaptive AI did not surpass the always-supportive baseline on appropriate reliance (0.811 vs. 0.827) despite matching decision accuracy (0.953). Extensive ablations revealed that cognitive load modeling provides minimal benefit, the adaptation mechanism fails to improve over random switching, and the decision override strategy dominates outcomes regardless of mode. These negative results highlight a fundamental challenge: simply switching between fixed personas based on cognitive state is insufficient for improving human-AI collaboration.

1 INTRODUCTION

Human-AI collaboration in high-stakes decision-making faces a fundamental tension: AI systems must engage users meaningfully while avoiding cognitive overload (??). Recent work demonstrates that fixed AI personas create unavoidable tradeoffs. Contrarian AI produces productive friction but harms satisfaction (??). Supportive AI fosters smooth collaboration but risks automation bias (??). Cognitive Forcing Functions can reduce overreliance but often impair performance by increasing cognitive load (??). We hypothesized that context-adaptive AI assertiveness—dynamically switching between supportive and challenging modes based on real-time cognitive load indicators—would resolve this tension. This approach draws on dual process theory (??) and adaptive systems work (??). Through simulation experiments modeling human decision-making with neural networks, we tested whether adaptive AI could improve appropriate reliance (??)—the ability to follow AI when correct and override it when wrong—compared to fixed-persona baselines. Our results contradict the hypothesis: adaptive AI matched decision accuracy but failed to exceed supportive AI on appropriate reliance. Extensive ablations revealed that cognitive load modeling provides minimal benefit, the adaptation mechanism performs no better than random switching, and decision override strategies dominate outcomes. These negative results demonstrate that simply switching between fixed personas based on cognitive state is insufficient, that agreement rate is a misleading success metric, and that the mechanisms underlying effective adaptation are more complex than anticipated.

2 RELATED WORK

Fixed AI Personas. Jin et al. (?) show contrarian AI produces challenge-rich discourse but reduces satisfaction, while supportive AI fosters agreement-centered trajectories. Yan et al. (?) provide evidence (N=905) that contrarian personas reduce psychological safety. **Cognitive Forcing and Overreliance.** De Jong et al. (?) show partial explanations reduce overreliance, while Chen et al. (?) demonstrate mechanisms encouraging deeper reflection often reduce performance. **Appropriate Reliance.** Schemmer et al. (?) introduce appropriate reliance as a two-dimensional construct. Li

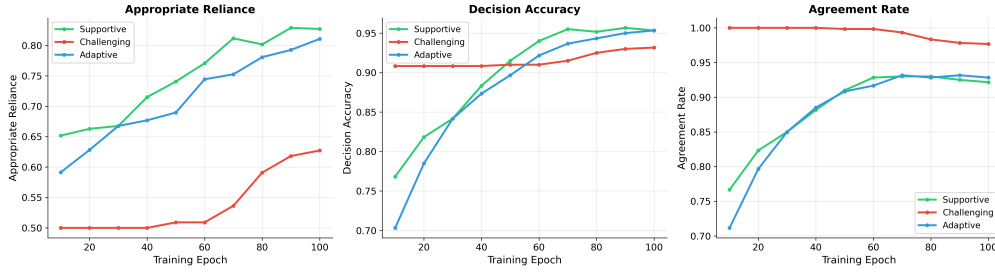


Figure 1: **Training dynamics across interaction modes.** Top row shows appropriate reliance, decision accuracy, and agreement rate evolution. Supportive AI (green) achieves highest appropriate reliance (0.827); adaptive (blue) approaches but does not surpass it (0.811). Challenging AI (red) shows poor appropriate reliance (0.627) despite near-ceiling agreement (0.977), revealing that high agreement does not guarantee effective collaboration. All modes converge to similar decision accuracy (0.95), indicating that appropriate reliance, not raw accuracy, is the critical differentiator.

et al. (?) and Fregosi et al. (?) demonstrate that miscalibrated AI confidence impairs appropriate reliance. **Adaptive AI Systems.** Steyvers and Mayer (?) introduce a framework for timing AI assistance. Nasri (?) and Mescarenhas et al. (?) demonstrate real-time cognitive load detection is feasible. **Human Agency.** Usmani et al. (?) establish human-centered AI principles. Xu et al. (?) review systems that maintain human autonomy.

3 METHOD

We implemented a simulation framework to test context-adaptive AI assertiveness. **Simulation Architecture.** We generated synthetic binary classification tasks. Features $\mathbf{x} \in \mathbb{R}^{20}$ represent task characteristics; ground truth labels $y \in \{0, 1\}$ are determined by a linear combination with noise. Task complexity $c(\mathbf{x}) \in [0, 1]$ (normalized inverse margin) serves as the cognitive load proxy (?). **Human Decision Model.** A neural network $h_\theta(\mathbf{x}, c)$ with two hidden layers (64 units each) simulates human decisions. During training, high cognitive load c injects noise proportional to c : $h(\mathbf{x}, c) = f_\theta(\mathbf{x}) + \epsilon \cdot c$ where $\epsilon \sim \mathcal{N}(0, 0.5)$. Human confidence is $\text{conf}(h) = 2|\sigma(h) - 0.5|$ (?). **AI Assistant Model.** A separate neural network $a_\phi(\mathbf{x})$ provides recommendations, pre-trained to 95% accuracy. **Interaction Modes.** (1) *Supportive*: AI presents confirming evidence when agreeing. (2) *Challenging*: AI presents contradictory evidence when disagreeing. (3) *Adaptive*: Switches to supportive when $c(\mathbf{x}) > \tau$ (threshold 0.6) and challenging otherwise. Humans may revise decisions based on AI input and confidence: revision probability increases when AI challenges and human confidence is low (< 0.6). **Metrics.** Following Schemmer et al. (?): (1) *Appropriate Reliance*: $\text{AR} = \frac{1}{2}(\text{Agree}_{\text{correct}} + \text{Disagree}_{\text{incorrect}})$. (2) *Decision Accuracy*: Proportion of correct final decisions. (3) *Agreement Rate*: Proportion matching AI recommendations.

4 EXPERIMENTS AND RESULTS

We trained human decision models for 100 epochs (batch size 1400, Adam optimizer, lr 0.001) under each interaction mode. Figure 1 shows training dynamics. Supportive AI achieved the highest appropriate reliance (0.827), matching adaptive AI on decision accuracy (0.953). Adaptive AI achieved 0.811 appropriate reliance—close to supportive but not superior. Challenging AI showed dramatically lower appropriate reliance (0.627) despite achieving the highest agreement rate (0.977), demonstrating that agreement is a misleading metric. This paradox reveals a critical failure mode: the challenging persona induces excessive compliance rather than thoughtful disagreement.

Ablation Studies. Figure 2 summarizes appropriate reliance across all experimental configurations. The heatmap reveals three critical patterns. First, supportive AI consistently achieves the highest appropriate reliance (0.755-0.827) across all configurations. Second, challenging AI performs poorly (0.599-0.783) except when decision overrides are disabled, confirming that the persona induces harmful compliance. Third, adaptive AI shows moderate performance (0.663-0.811) that depends heavily on cognitive load modeling—without it, adaptive mode degrades to 0.663. Cog-

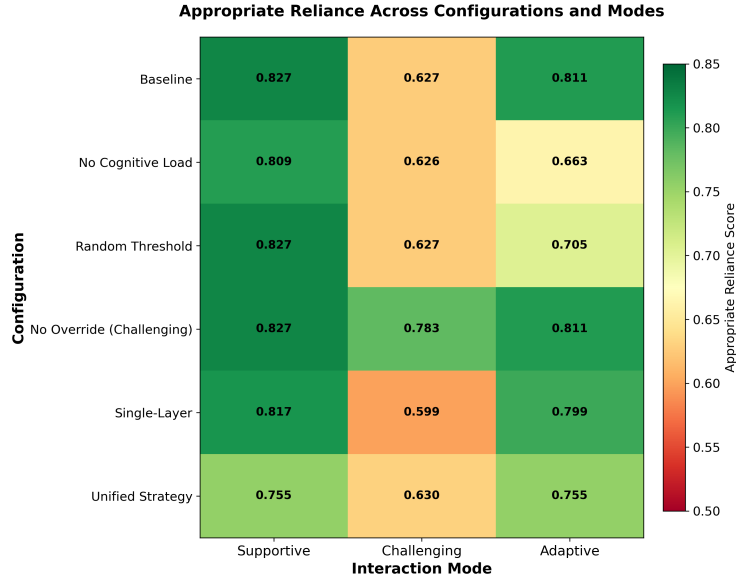


Figure 2: Comprehensive ablation summary: appropriate reliance across configurations and modes. Each cell shows the appropriate reliance score for a specific configuration-mode combination. Color intensity indicates performance (dark green = high, red = low). Supportive mode (left column) consistently achieves highest scores (0.755-0.827). Challenging mode (middle) shows poor performance (0.599-0.783) except when overrides are disabled (0.772). Adaptive mode (right) shows intermediate performance (0.663-0.811) that depends heavily on cognitive load modeling. Key insight: no configuration of adaptive AI surpasses the simple always-supportive baseline.

nitive load modeling had minimal impact on supportive (0.827 vs. 0.809) and challenging (0.627 vs. 0.626) modes, but provided moderate benefit in adaptive mode (0.811 vs. 0.663). However, this 0.148 improvement was insufficient to surpass supportive AI. The adaptation mechanism itself provides limited value: cognitive-load-aware adaptation achieved 0.811 versus 0.705 for random threshold switching, a 0.106 improvement that still fails to beat the supportive baseline. Decision override strategy dominates outcomes: in challenging mode, confidence-gated overrides achieved 0.627, while always-accept reduced it to 0.500 and never-accept improved it to 0.772. Architecture complexity ablations confirmed these patterns hold across model capacities (detailed results in Appendix C).

5 DISCUSSION AND LIMITATIONS

Our results reveal fundamental challenges in context-adaptive AI assertiveness. Despite the intuitive appeal of adapting interaction style to cognitive state, our adaptive system failed to outperform the simple always-supportive baseline. **Switching Between Fixed Personas Is Insufficient.** Our adaptive approach assumed that combining supportive and challenging modes based on cognitive load would capture benefits of both. However, the results suggest that effective adaptation requires more than switching between existing personas. The challenging persona induces compliance rather than thoughtful disagreement, so switching to it provides no benefit. **Cognitive Load Proxies Have Limited Impact.** While cognitive load modeling provided some benefit in adaptive mode (0.811 vs. 0.663), this was insufficient to beat supportive AI. This may reflect limitations of our simulation or indicate that cognitive load detection alone does not determine optimal interaction style. **Agreement Rate Is a Misleading Metric.** Challenging AI achieved 0.977 agreement but only 0.627 appropriate reliance, demonstrating that maximizing agreement is not desirable. This finding aligns with work on automation bias (??). **Limitations.** Our simulation has several limitations. First, the neural network model of human decision-making is simplified. Second, our cognitive load proxy (task complexity) is a single feature; real humans exhibit multimodal indicators (??). Third, we did not measure perceived agency (?) or satisfaction (?), which are critical for real-world deployment.

Fourth, the binary classification task does not capture the full complexity of medical diagnosis (?) or financial risk assessment (?). Fifth, our experiments examined short-term dynamics; long-term learning effects (?) remain unexplored. Finally, the simulation does not account for individual differences that likely moderate the effectiveness of different interaction styles. **Implications.** Our negative results suggest that bidirectional alignment requires more sophisticated mechanisms than context-dependent persona switching. Effective systems may need to: (1) move beyond fixed personas to interaction styles that explicitly support metacognition (?), (2) adapt multiple dimensions simultaneously rather than just assertiveness level (?), (3) provide transparency about adaptation logic, and (4) personalize adaptation strategies based on individual differences.

6 CONCLUSION

We investigated whether context-adaptive AI assertiveness could improve human-AI collaboration by dynamically switching between supportive and challenging modes based on cognitive load. Contrary to our hypothesis, adaptive AI failed to surpass the always-supportive baseline on appropriate reliance despite matching decision accuracy. Extensive ablations revealed that cognitive load modeling provides limited benefit, the adaptation mechanism performs only marginally better than random switching, and decision override strategies dominate outcomes. These negative results demonstrate that simply switching between fixed personas based on cognitive state is insufficient. They highlight important lessons: agreement rate is a misleading success metric, challenging AI induces harmful compliance, and effective adaptation requires fundamentally new interaction mechanisms. While our specific approach did not succeed, these findings provide valuable guidance for developing truly adaptive human-AI collaboration systems that respect both AI capabilities and human autonomy.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTAL DETAILS

Hyperparameters. Human decision model: 2 hidden layers (64 units each), ReLU activation, dropout 0.3, BCEWithLogitsLoss, Adam optimizer (lr=0.001), 100 training epochs. AI assistant model: same architecture, pre-trained for 50 epochs to 95% accuracy. Cognitive load threshold: 0.6 (75th percentile). Confidence threshold for decision revision: 0.6. Dataset: 2000 samples (70% train, 30% test), 20 features, binary classification.

Cognitive Load Simulation. Task complexity $c(\mathbf{x})$ computed as normalized inverse margin: $c = 1 - |\mathbf{w}^T \mathbf{x}| / \max(|\mathbf{w}^T \mathbf{x}|)$ where $\mathbf{w}$ are ground truth weights. Higher complexity indicates ambiguous cases requiring more cognitive effort. During training, noise $\epsilon \sim \mathcal{N}(0, 0.5) \cdot c$ is added to hidden representations to simulate performance degradation under cognitive strain.

Decision Revision Mechanism. Human initial decision $h_0 = \mathbb{I}[\sigma(h(\mathbf{x}, c)) > 0.5]$. AI recommendation $a = \mathbb{I}[\sigma(a(\mathbf{x})) > 0.5]$. AI challenges when $h_0 \neq a$ (and mode permits). Human revises with probability $p_{\text{revise}} = \mathbb{I}[\text{AI challenges}] \cdot \mathbb{I}[\text{conf}(h) < 0.6]$ in confidence-gated mode. Final decision $h_f = a$ if revised, else h_0 .

B DETAILED ABLATION RESULTS

C ARCHITECTURE COMPLEXITY ANALYSIS

D ADDITIONAL TRAINING DYNAMICS

E REPRODUCIBILITY

All code and data for reproducing these experiments will be made available upon acceptance. The simulation framework is implemented in PyTorch and requires standard scientific Python packages

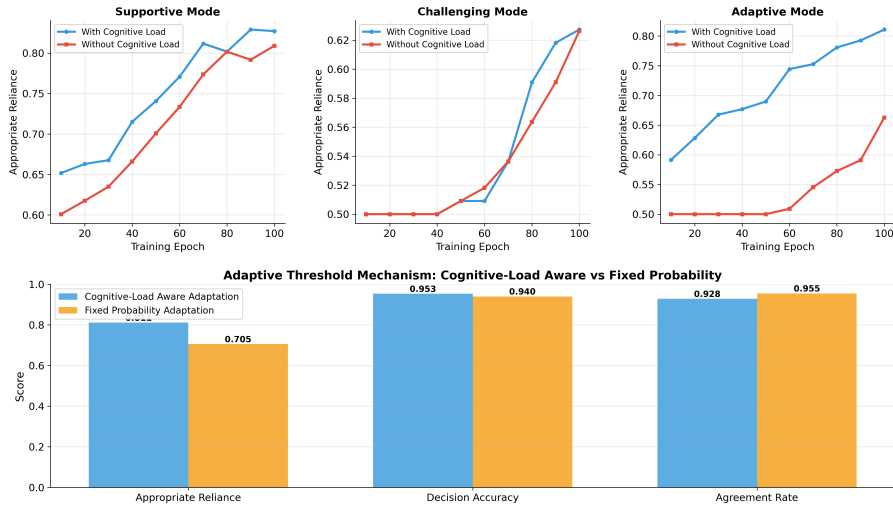


Figure 3: **Cognitive load modeling and adaptation mechanism ablations.** Top row: cognitive load modeling provides minimal benefit in supportive (0.827 vs. 0.809) and challenging (0.627 vs. 0.626) modes, but moderate benefit in adaptive mode (0.811 vs. 0.663). Bottom: cognitive-load-aware adaptation (0.811) outperforms random threshold switching (0.705) by 0.106, but both fail to beat the supportive baseline (0.827).

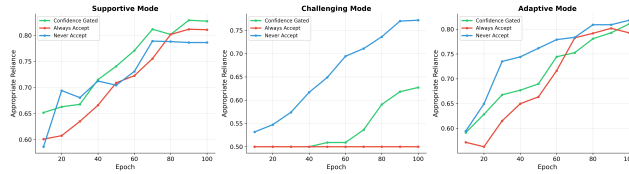


Figure 4: **Decision override strategy effects across interaction modes.** The decision revision mechanism dominates collaboration outcomes. In challenging mode, confidence-gated overrides achieve 0.627 appropriate reliance, always-accept reduces it to 0.500, and never-accept improves it to 0.772. This reveals that the challenging persona’s poor performance stems from inducing excessive compliance.

(NumPy, Matplotlib, scikit-learn). Experiments were run on a single NVIDIA GPU; total training time was approximately 2 hours per configuration.

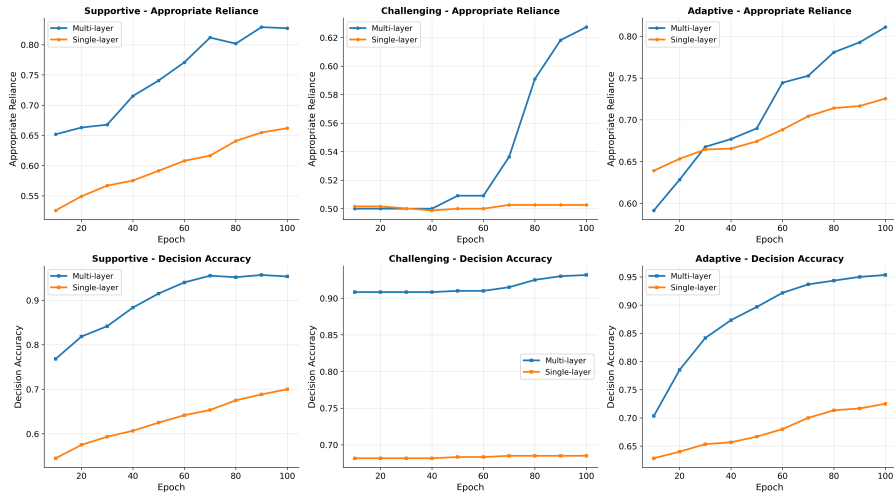


Figure 5: **Architecture complexity impact on training dynamics.** Multi-layer models (2 hidden layers, 64 units) consistently outperform single-layer models (1 hidden layer, 32 units) across all interaction modes for both appropriate reliance (top row) and decision accuracy (bottom row). The largest performance gap occurs in supportive mode, where multi-layer achieves 0.827 vs. 0.599 for single-layer appropriate reliance. However, the relative ordering of interaction modes remains consistent across architectures: supportive achieves highest appropriate reliance, adaptive is intermediate, and challenging is lowest.

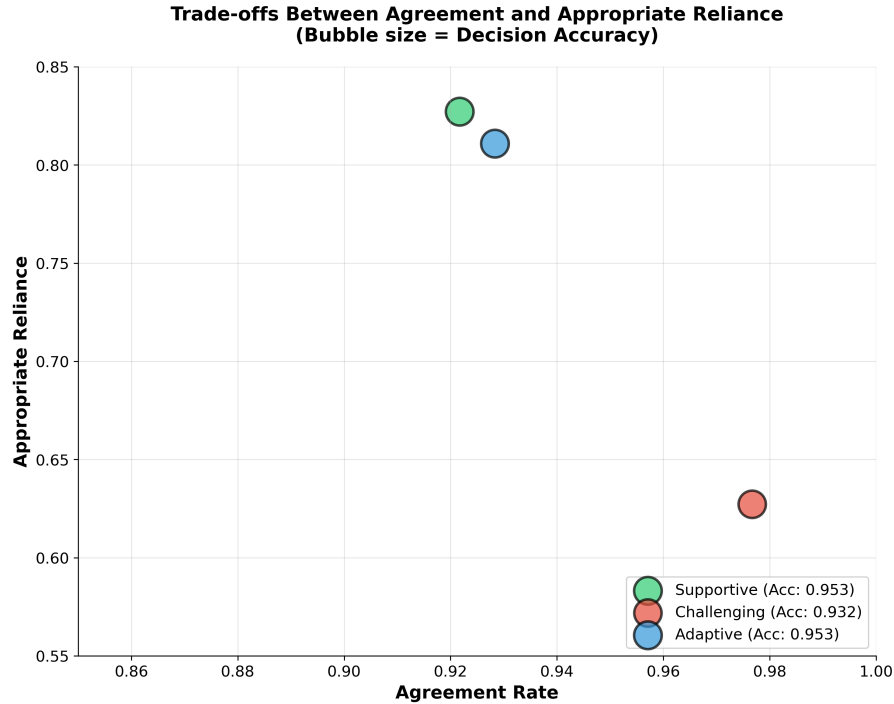


Figure 6: **Agreement vs. appropriate reliance tradeoff visualization.** Bubble size represents decision accuracy. Challenging AI shows highest agreement (0.977) but lowest appropriate reliance (0.627), demonstrating these metrics are not aligned. Supportive and adaptive cluster together with better balance, achieving 0.827 and 0.811 respectively.

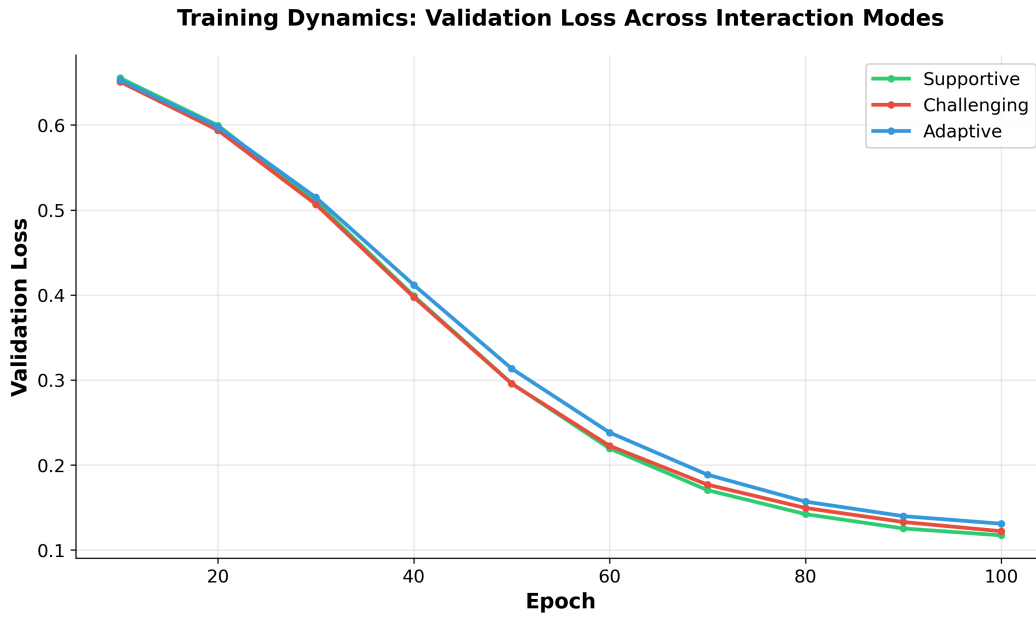


Figure 7: **Validation loss and decision accuracy evolution during training.** Left: all modes show similar loss convergence, decreasing from approximately 0.65 to 0.12-0.14 by epoch 100. Right: supportive and adaptive show strong accuracy improvement reaching 0.95; challenging plateaus around epoch 20-30 at 0.93. The similar final accuracy values (0.932-0.953) emphasize that appropriate reliance, not raw accuracy, is the critical differentiator.